



Mahidol University
Wisdom of the Land

Pharmacokinetics

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เอกสารประกอบการสอนรายวิชาเภสัชวิทยา 1 (ภกปว 304) สำหรับนักศึกษาเภสัชศาสตร์ชั้นปีที่ 3

บรรยายวันที่ 7 สิงหาคม 2568 (8.00-10.00 น.) ห้อง T203 ชั้น 2 อาคารเทพรัตน์

จัดทำวันที่ 10 สิงหาคม 2563 และปรับปรุงวันที่ 5 สิงหาคม 2568



Learning Objectives

After the class, students should be able to

1. Explain the pharmacokinetic processes, including
 - **A**bsorption
 - **D**istribution
 - **M**etabolism
 - **E**xcretion
2. Explain any factor that may influence the pharmacokinetic processes
3. Explain the pharmacokinetic parameters



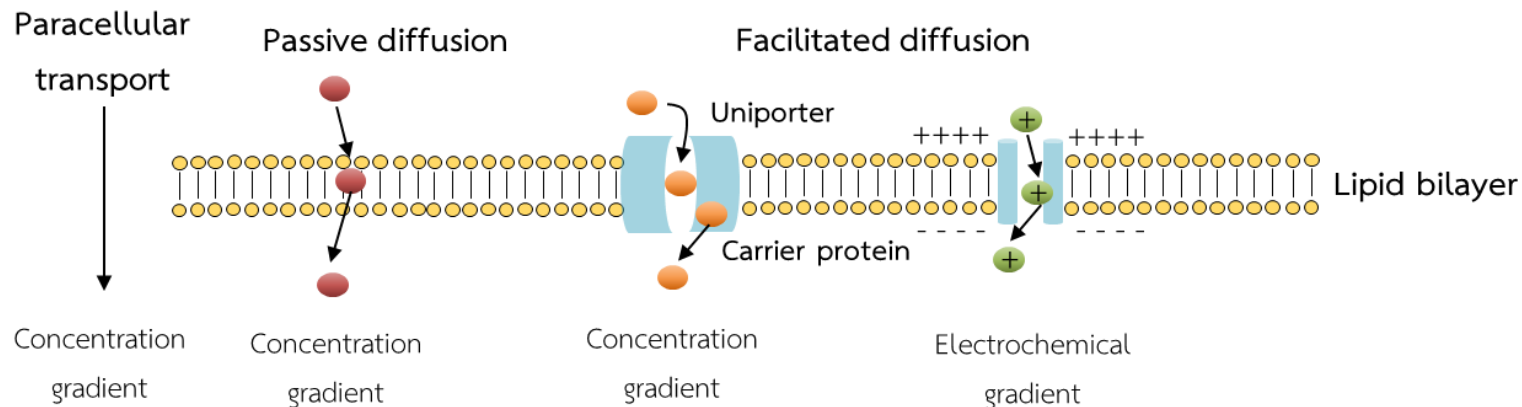
Introduction to pharmacokinetics

- **Pharmacokinetics** (*kinētikos* = motion) is defined as
 - The movement of drug within the body
 - Also refer to as “what the body does to a drug”
- **Pharmacokinetic processes**
 1. **A**bsorption
 2. **D**istribution
 3. **M**etabolism
 4. **E**xcretion

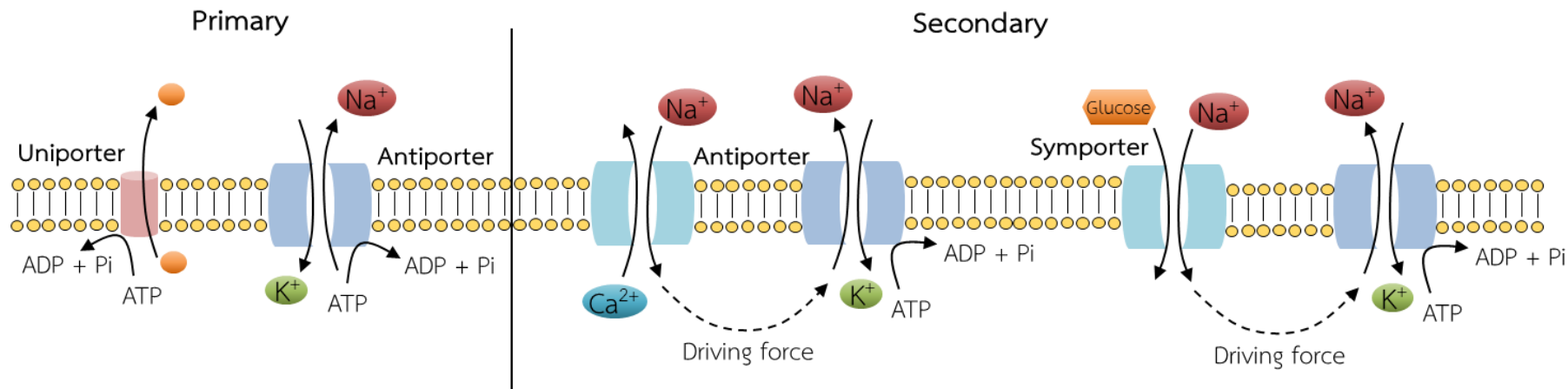
These processes involve the passage of drug through the membrane

Passage of drugs across membrane barriers

Passive transport



Active transport





Passive Diffusion

- Transmembrane movement of most drugs
- Diffusion of drug along a concentration gradient by virtue of its solubility in the lipid bilayer
 - Lipid:water partition coefficient of the drug
 - Membrane surface area exposed to the drug
- At steady state
 - **Nonelectrolyte drugs**: the concentration of the unbound drug is the same on both sides of the membrane
 - **Ionic compounds**: depends on the electrochemical gradient for the ion and on differences in pH across the membrane
 - Ion trapping on one side of the membrane



Influence of pH on drug ionization

- Many drugs are weak acids or bases
- Present in solution as both
 - Nonionized form: the lipid-soluble, diffusible
 - Ionized form: relatively lipid insoluble, poorly diffusible
 - **Carboxylic acids**
 - **Amino groups**
 - pK_a = the pH at which half the drug is in its ionized form
- Henderson-Hasselbalch equation

$$pH = pK_a - \log \frac{[\text{protonated form}]}{[\text{unprotonated form}]}$$

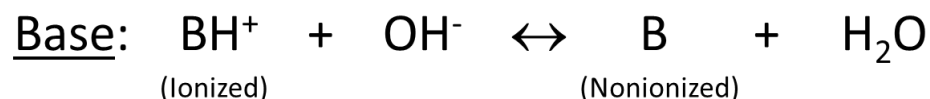
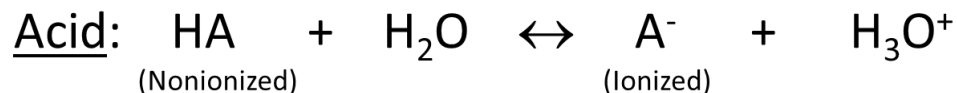
Example calculation

Henderson-Hasselbalch equation:

$$\log \frac{\text{(Protonated)}}{\text{(Unprotonated)}} = \text{pKa} - \text{pH}$$

Protonated

Unprotonated



- Example: Diclofenac (weak acid, pKa 3.9)

$$\log \frac{\text{(Protonated)}}{\text{(Unprotonated)}} = 3.9 - 7.4$$

(Physiologic pH)

$$\frac{\text{(Protonated)}}{\text{(Unprotonated)}} = 10^{-3.5}$$

$$\frac{\text{(Unprotonated)}}{\text{(Protonated)}} = 10^{3.5}$$

(Ionized form)

$$\log \frac{\text{(Protonated)}}{\text{(Unprotonated)}} = 3.9 - 6.4$$

(pH at inflamed tissue)

$$\frac{\text{(Protonated)}}{\text{(Unprotonated)}} = 10^{-2.5}$$

$$\frac{\text{(Unprotonated)}}{\text{(Protonated)}} = 10^{2.5}$$

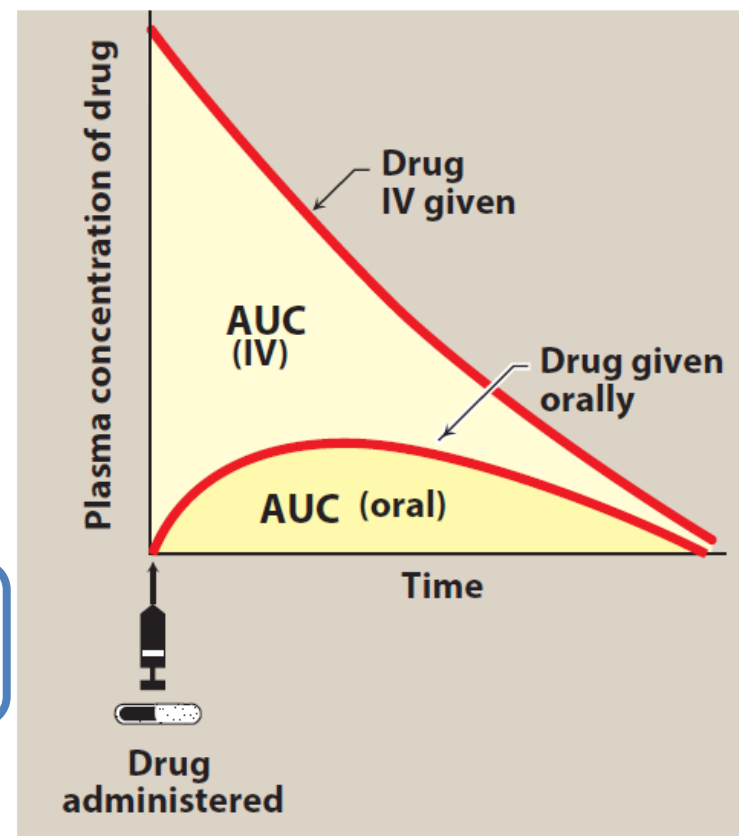
1. Drug Absorption

- The movement of a drug from its site of administration into the central compartment
- Bioavailability** describes the fractional extent to which an administered dose of drug reaches its site of action or a biological fluid (usually the systemic circulation)

$$F = \frac{\text{Quantity of drug reaching systemic circulation}}{\text{Quantity of drug administered}}$$

– where $0 < F \leq 1$ (or $0\% < F \leq 100\%$)

$$\text{Relative bioavailability} = \frac{\text{AUC studied route/administered dose}}{\text{AUC IV route/administered dose}}$$





Drug Absorption (Cont.)

- Most **drug absorption from the GI tract** occurs by **passive diffusion**
- Absorption is favored when the drug is in nonionized, more lipophilic form
 - **Weak acids would be better absorbed from the stomach (pH 1–2) than from the upper intestine (pH 3–6)**
 - **Weak bases, vice versa**
- Absorption from the GI tract is governed by factors such as
 - Surface area for absorption (area of intestine is larger than stomach)
 - Blood flow to the site of absorption
 - Physical state of the drug (solution, suspension, or solid dosage form)
 - Its aqueous solubility
 - Drug's concentration at the site of absorption



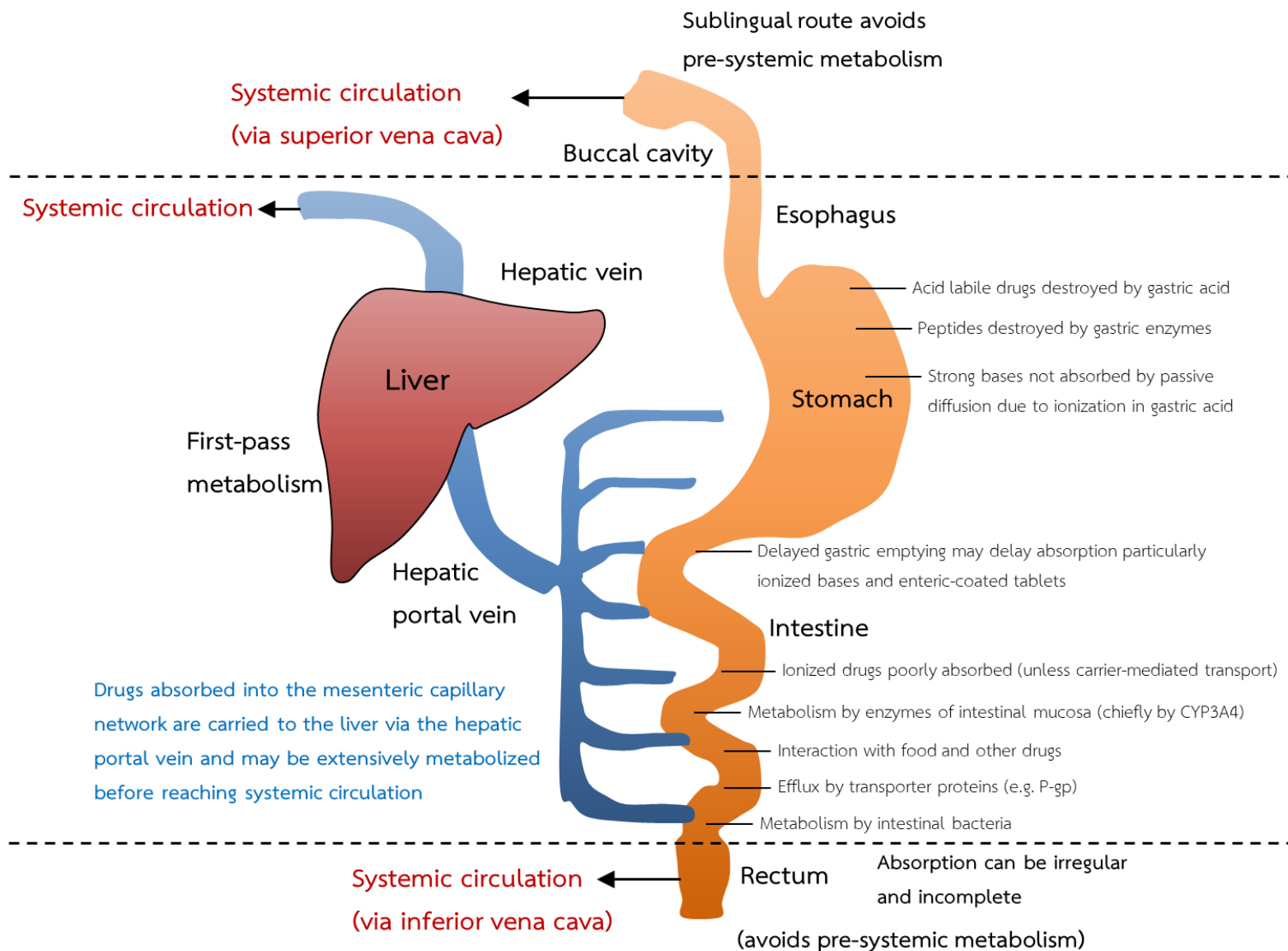
The influences of gastric emptying on drug absorption

- Any factor that accelerates gastric emptying will generally increase the rate of drug absorption
 - Any factor that delays gastric emptying will have the opposite effect
- Gastric emptying rate is influenced by numerous factors, including
 - The caloric content of food
 - Diurnal and interindividual variation
 - Metabolic state (rest or exercise)
 - Estrogen slows gastric emptying

The influences of dosage form on drug absorption

- Enteric coating
 - Prevents gastric irritation
 - But, prevents dissolution of acid labile drug in acidic gastric contents
- Controlled-release, extended-release, sustained-release, and prolonged-action
 - **Designed to produce slow, uniform absorption of the drug for ≥ 8 h**
 - Other potential advantages
 - Reduction in the frequency of administration
 - Maintenance of a therapeutic effect overnight
 - Decreased incidence and intensity of undesired effects (peaks activity)
 - Avoid nontherapeutic blood levels of the drug (trough concentration)

Absorption throughout the gastrointestinal tract





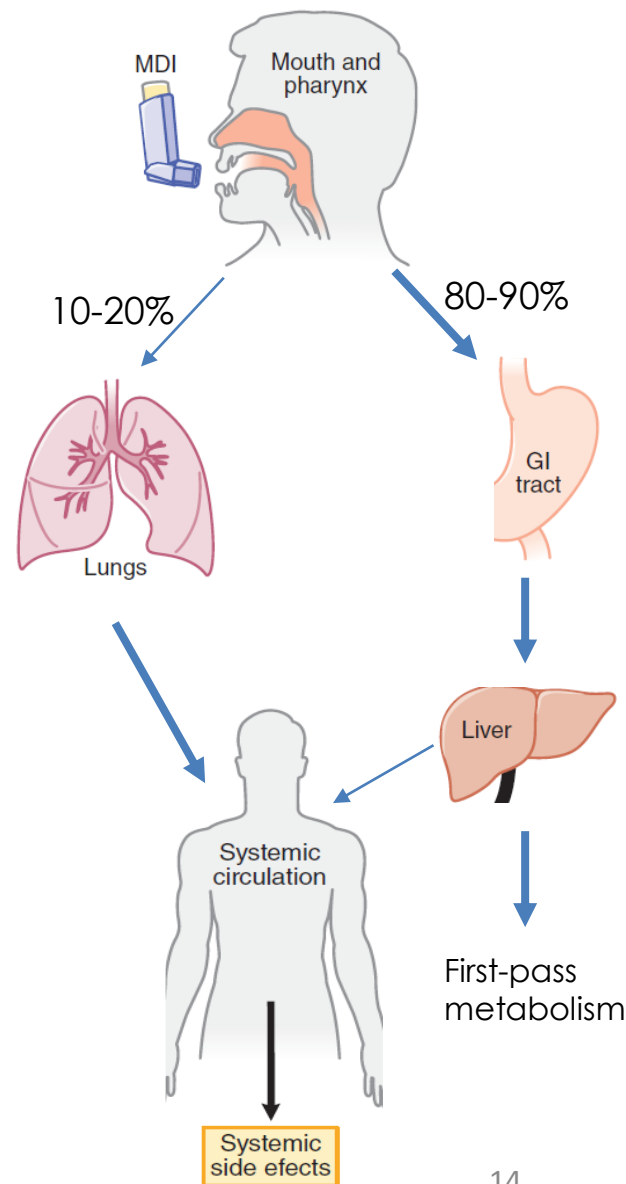
Absorption of parenteral routes

ROUTE AND BIOAVAILABILITY (F)	Absorption	PATTERN SPECIAL UTILITY	LIMITATIONS AND PRECAUTIONS
Intravenous F = 1 by definition	Absorption circumvented - Potentially immediate effects - Suitable for large volumes and for irritating substances, or complex mixtures, when diluted	Valuable for emergency use Permits titration of dosage - Usually required for high-molecular weight protein and peptide drugs	Increased risk of adverse effects - Must inject solutions slowly as a rule - Not suitable for oily solutions or poorly soluble substances
Subcutaneous $0.75 < F < 1$	Prompt from aqueous solution Slow and sustained from repository preparations	Suitable for some poorly soluble suspensions and for instillation of slow-release implants - Appropriate for self-administration (e.g., insulin)	Not suitable for large volumes - Possible pain or necrosis from irritating substances
Intramuscular $0.75 < F < 1$	Prompt from aqueous solution Slow and sustained from repository preparations	Suitable for moderate volumes, oily vehicles, and some irritating substances	Precluded during anticoagulant therapy - May interfere with interpretation of certain diagnostic tests (e.g., creatine kinase)
Oral ingestion $.05 < F < 1$	Variable, depends on many factors	Most convenient and economical; usually safer	Requires patient compliance Bioavailability potentially erratic and incomplete

Other routes of administration

- Pulmonary Absorption

- Gaseous and volatile drugs may be inhaled and absorbed through the pulmonary epithelium and mucous membranes of the respiratory tract
- Access to the circulation is rapid because the lung's surface area is large
- Advantages are: the almost instantaneous absorption of a drug into the blood, avoidance of first-pass loss, local application of the drug at the desired site



- Topical Application

- Mucous Membranes, Eye
- Skin: Transdermal Absorption
 - Absorption is dependent on the surface area and lipid solubility



2. Distribution of Drugs

- Cardiac output
 - [Regional blood flow](#)
 - Capillary permeability
 - [Tissue volume](#)
- } affect the rate of delivery and amount of drug distributed into tissues
- Initially, liver, kidney, brain, and other well-perfused organs receive most of the drug
 - Delivery to muscle, most viscera, skin, and fat is slower

[In 70 kg male;](#)

	KIDNEYS	HEART	LIVER	BRAIN	SKELETAL MUSCLE	FAT	REMAINDER	TOTAL
Blood Flow (mL/min)	1100	250	1700	800	900	250	500	5500
Mass (kg)	0.3	0.3	2.6	1.3	34	10	21.5	70
Flow/Mass (mL/min/kg)	3667	833	654	615	26	25	23	
% Cardiac Output	20	4.5	31	14.5	16.4	4.5	9.1	100



The influences of protein binding on drug distribution

- Many drugs circulate in the bloodstream bound to plasma proteins
 - The binding is usually reversible
 - Albumin binds acidic drugs; α 1-acid glycoprotein binds basic drugs
 - Others: hormone carrier proteins, e.g. sex hormone-binding globulin
- **Binding to plasma proteins limits**
 - **Drug concentration at its site of action** because only unbound drug is in equilibrium across membranes
 - **Drug's glomerular filtration, drug transport and metabolism**
- **Competition for plasma protein-binding may cause one drug to elevate the concentration of one bound less avidly**

Volume of distribution

- Volume of distribution (Vd) is an important indicator of the extent of drug distribution into body fluids and tissues

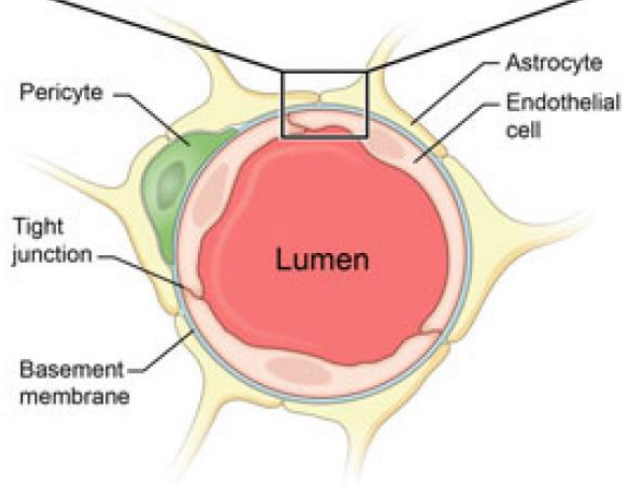
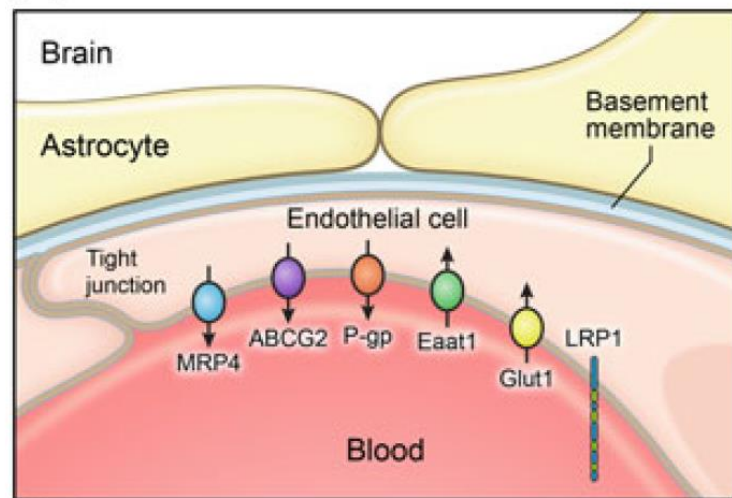
$$\text{volume of distribution} = \frac{\text{amount of drug}}{\text{concentration}}$$

In 70 kg male;

Body compartments			
	Plasma	Interstitial	Intracellular
Volume	3 L	9 L	29 L
Drugs/ chemicals	— Evan blue (Vd = 3 L) — Iodine 131-albumin (Vd = 3 L)	— Mannitol (Vd = 12 L) — Amoxicillin (Vd = 15 L) — Na ⁺ (Vd = 18 L)	— Enalapril (Vd = 40 L) — Urea (Vd = 41 L) — Lidocaine (Vd = 77 L) — Tetracycline (Vd = 100 L) — Atropine (Vd = 120 L) — Meperidine (Vd 300 L) — Chlorpromazine (Vd = 1,500 L) — Propofol (Vd = 4,000 L) — Chloroquine (Vd = 13,000 L)
	Small Vd		
	Medium Vd		
	Large Vd		
% Body weight	4	13	41

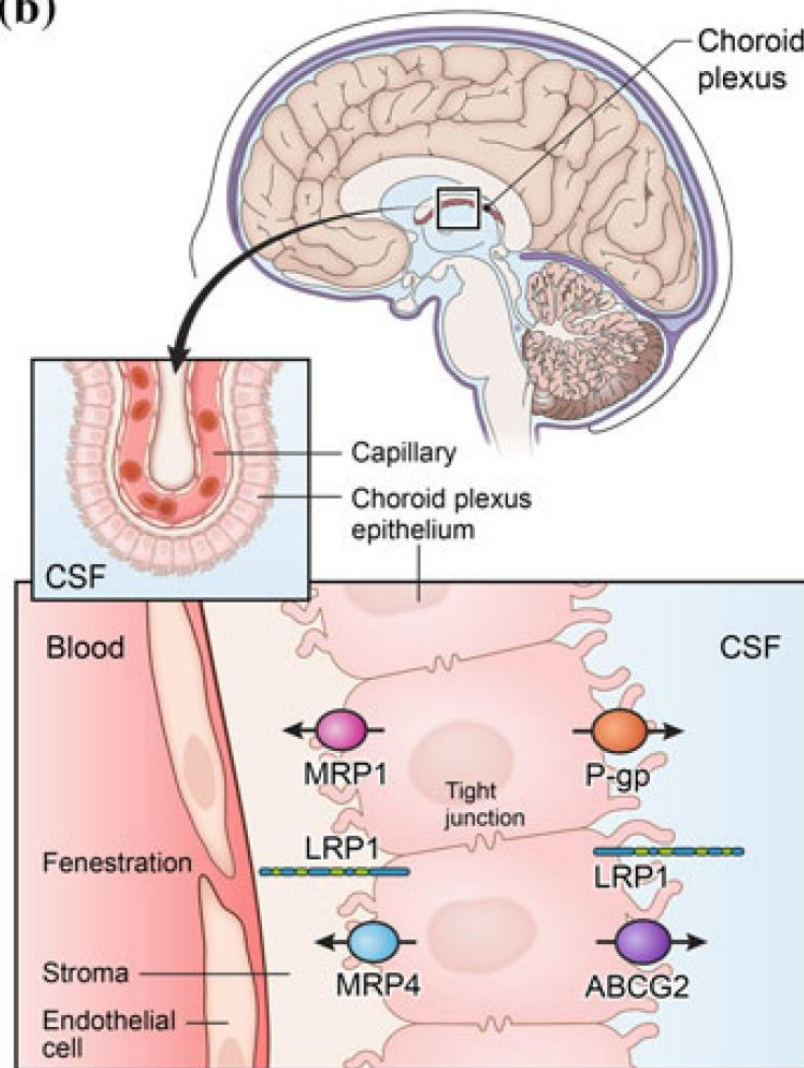
Blood-brain barrier and blood-CSF barrier

(a)



Blood-brain barrier

(b)



Blood-CSF barrier

Drug accumulation

- Many drugs accumulate in tissues at higher concentrations
- Tissue binding of drugs usually occurs with cellular constituents such as proteins, phospholipids, or nuclear proteins
 - Generally is reversible
- May serve as a reservoir that prolongs drug action in that same tissue or at a distant site reached through the circulation
 - e.g., macrolides and respiratory fluoroquinolone antibiotics
- Tissue binding and accumulation also can produce local toxicity
 - e.g., renal and ototoxicity associated with aminoglycoside antibiotics

Drug accumulation (Cont.)

- **Bone**

- The tetracycline antibiotics (and other divalent metal-ion chelating agents) and heavy metals
- May accumulate in bone by adsorption onto the bone crystal surface and eventual incorporation into the crystal lattice

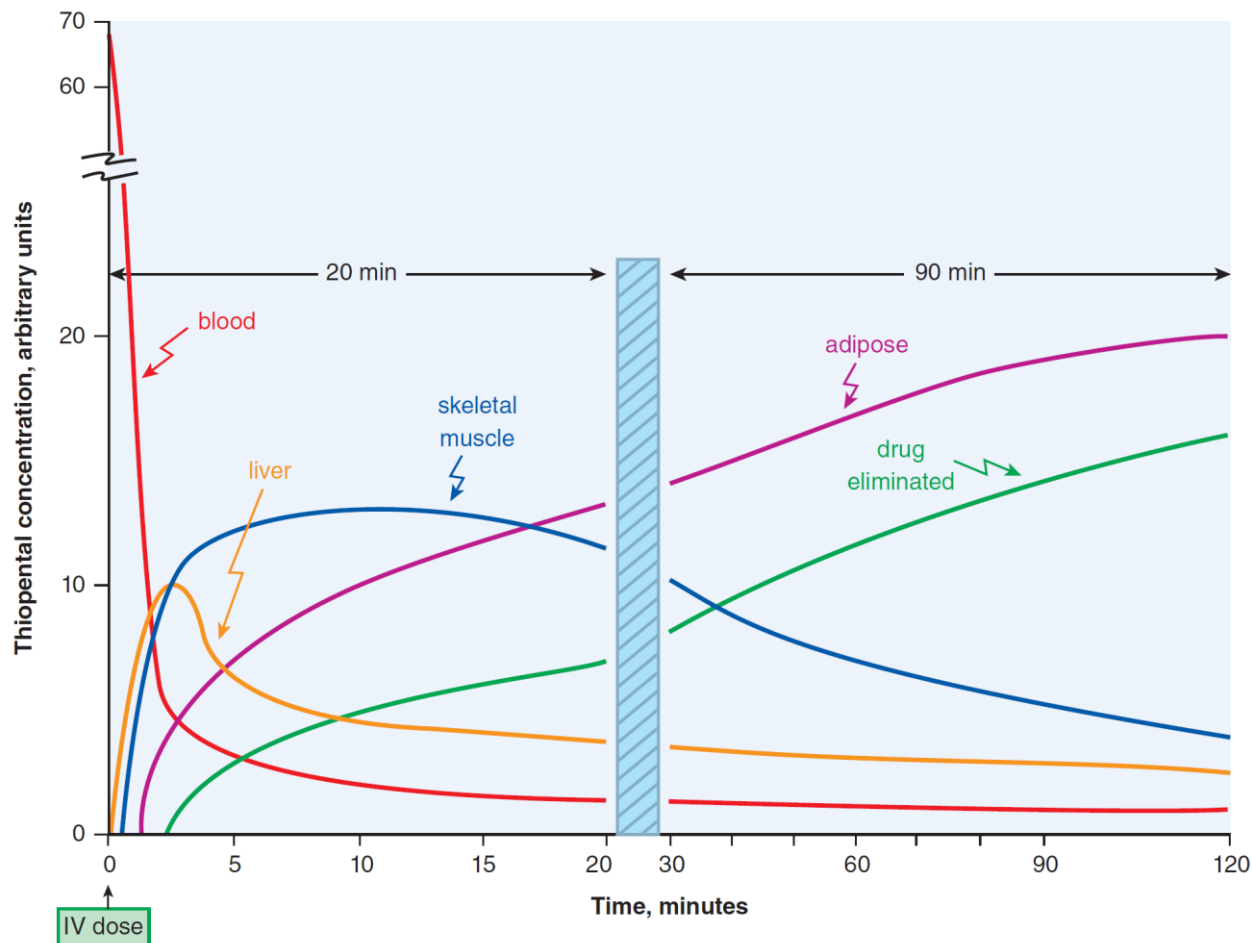


M Dent J 2014;34(1):55-69.

- **Fat**

- May serve as a reservoir for lipid-soluble drugs
- In obese persons, the fat content of the body may be as high as 50%

Drug redistribution



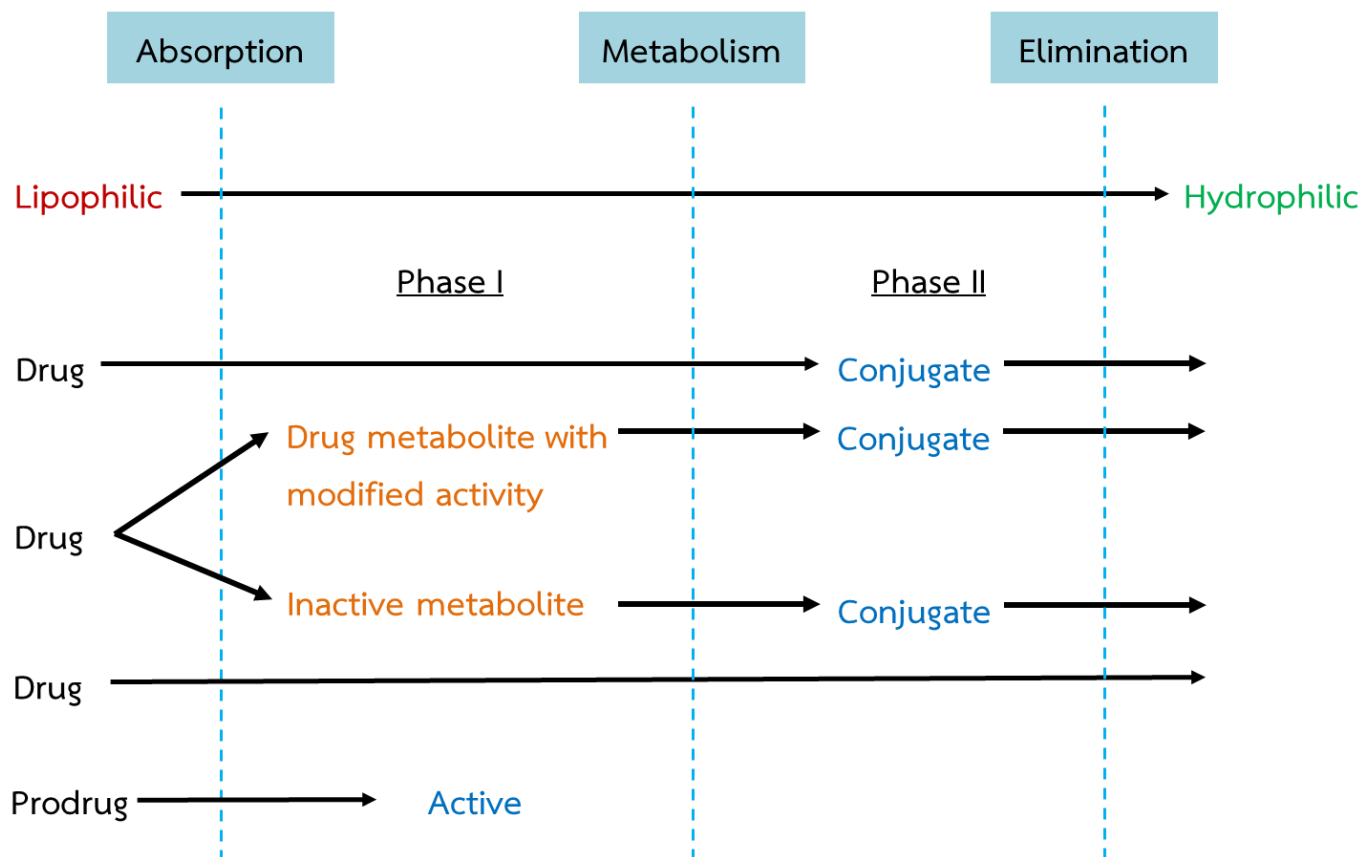
- IV anesthetic thiopental, a lipid-soluble drug
- Because blood flow to the brain is high and thiopental readily crosses the BBB
 - Reaches maximal concentration in brain rapidly
- Subsequently, the plasma and brain concentrations decrease as thiopental redistributes to other tissues, such as muscle and, finally, adipose tissue

Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 13e; 2018.

- QUICK ON-OFF

3. Metabolism of Drugs

- Biotransformation of drugs and other xenobiotics into more hydrophilic metabolites
 - Essential for renal elimination from the body, as well as for termination of the biological and pharmacological activity

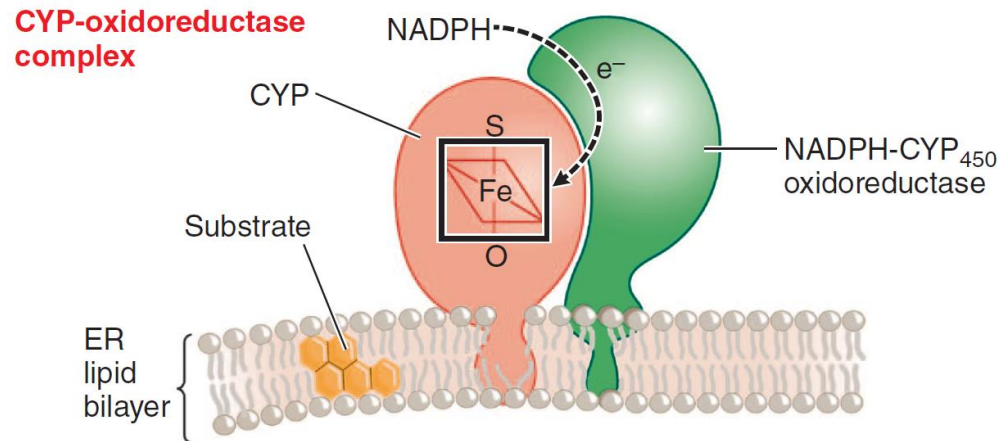
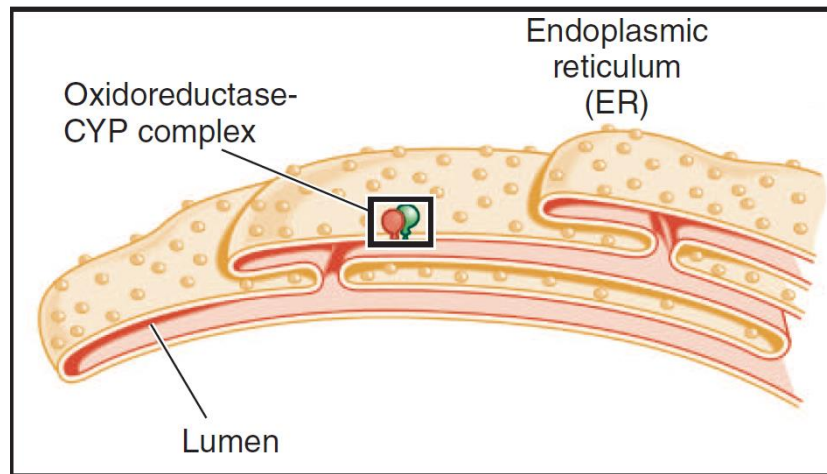




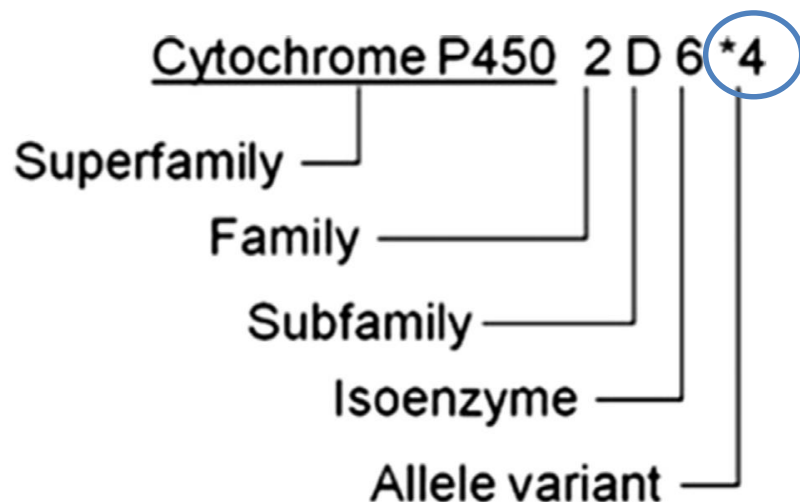
Drug-metabolizing enzymes

Enzymes	Reactions	Location
Phase I		
• Cytochrome P450s (P450 or CYP)	C and O oxidation, dealkylation, others	Microsome
• Flavin-containing monooxygenases (FMOs)	N, S, and P oxidation	Microsome
• Epoxide hydrolases (EHs)	Hydrolysis of epoxides	Microsome, Cytosol
Phases II “transferases”		
• UDP-glucuronosyltransferases (UGTs)	Addition of glucuronic acid	Microsome
• Sulfotransferases (SULT)	Addition of sulfate	Cytosol
• Glutathione-S-transferases (GSTs)	Addition of glutathione	Cytosol, Microsome
• N-Acetyltransferases (NATs)	Addition of acetyl group	Cytosol
• Methyltransferases (MTs)	Addition of methyl group	Cytosol
Other enzymes		
• Alcohol dehydrogenases	Reduction of alcohols	Cytosol
• Aldehyde dehydrogenases	Reduction of aldehydes	Cytosol
• NADPH-quinone oxidoreductase (NQO)	Reduction of quinones	Cytosol

Cytochrome P450 (CYP)



CYP Nomenclature:

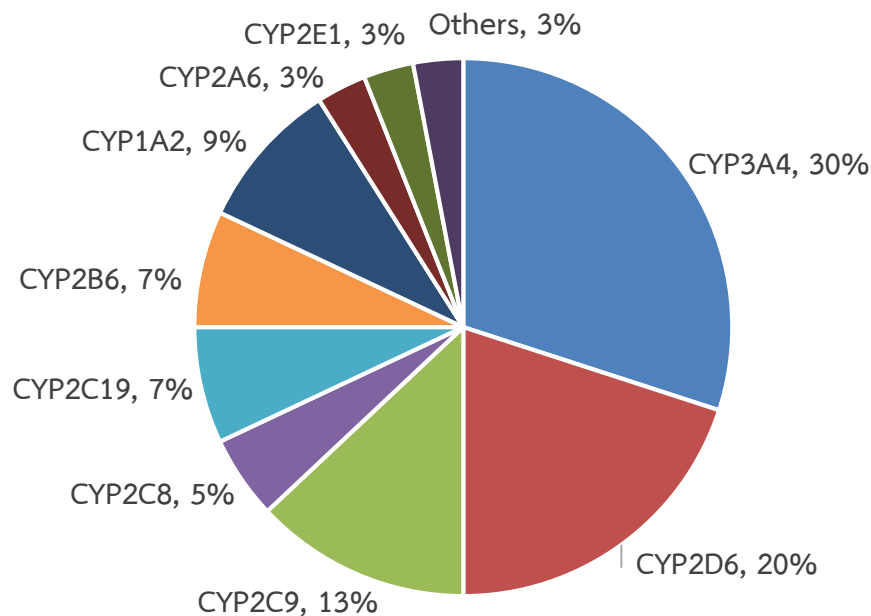


• CYP polymorphisms

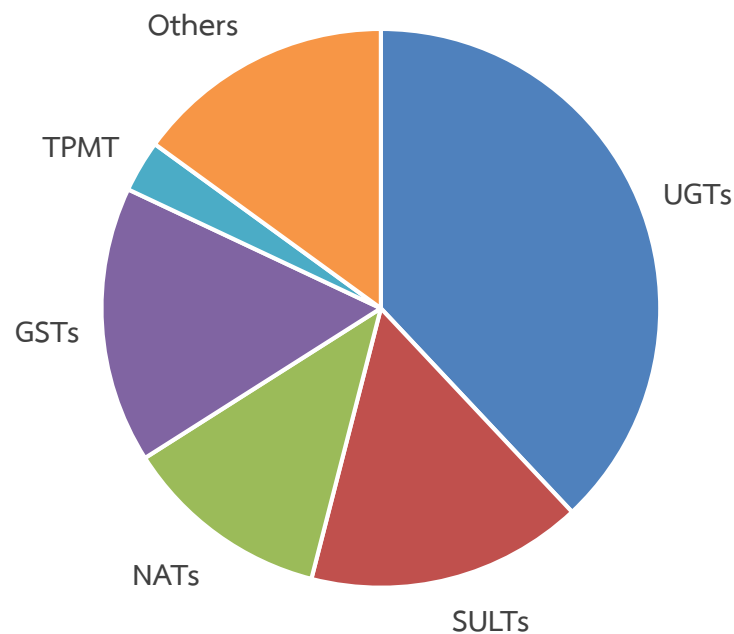
- Poor metabolizer
- Rapid metabolizer
- Ultra rapid metabolizer

Drug-metabolizing enzymes (Cont.)

Phase I



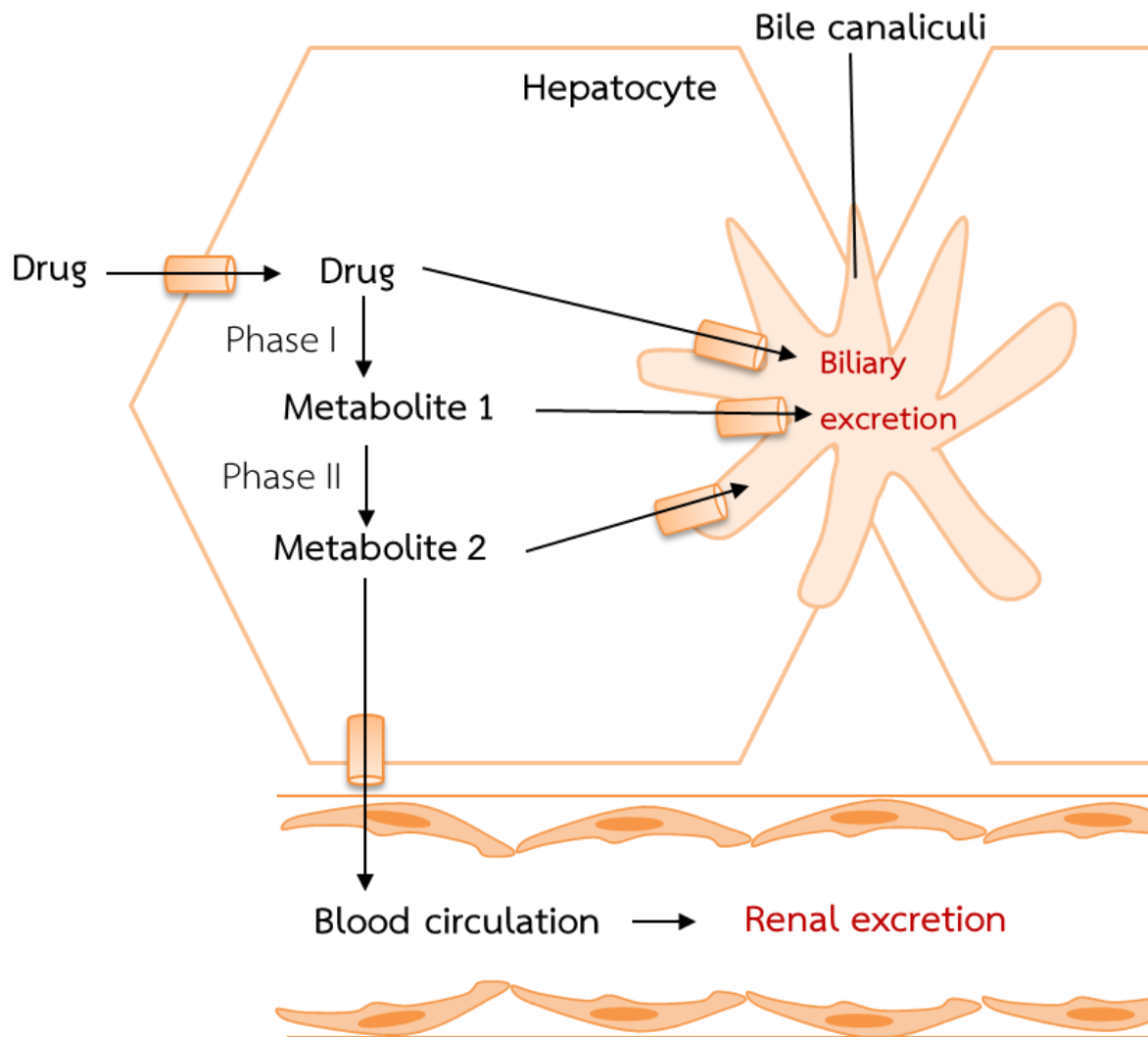
Phase II



Adapted from: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 13e; 2018.

- **Enzyme inducers/inhibitors**
- **Hepatic impairment may lead to a reduction in drug metabolism**
 - ↑ Aspartate aminotransferase (AST)
 - ↑ Alanine aminotransferase (ALT)

Export of drug and its metabolite from the liver

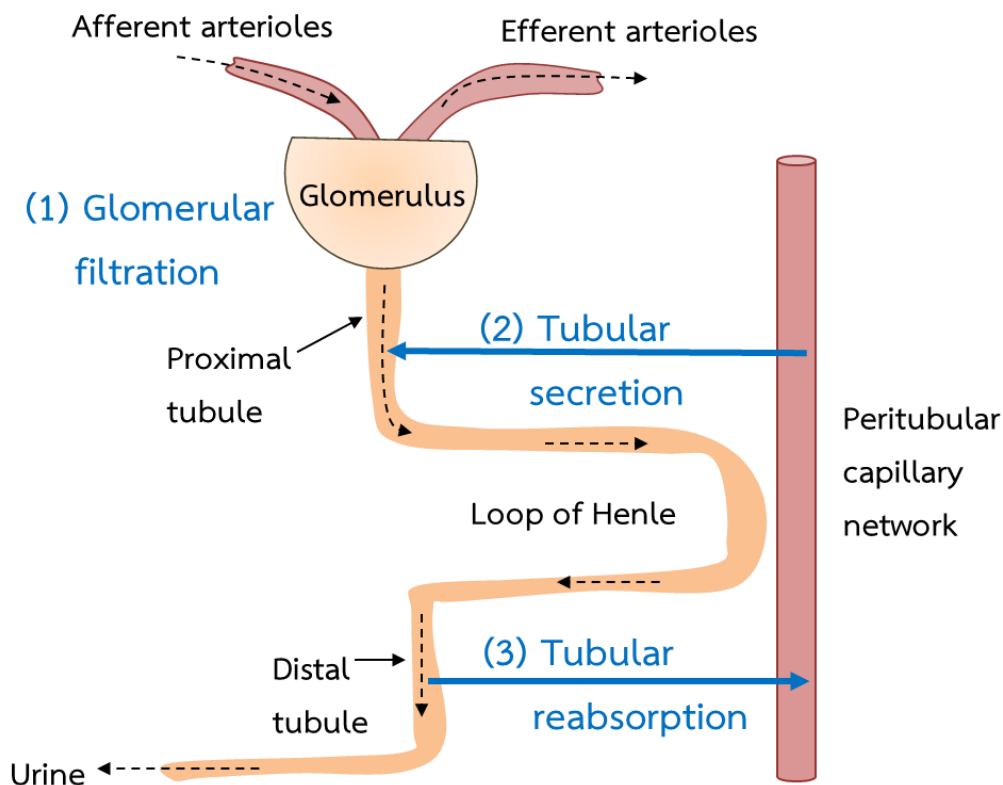




4. Excretion of Drugs

- Drugs are eliminated from the body either unchanged or as metabolites
 - 1) The kidney is the most important organ for excreting drugs and their metabolites
 - Renal excretion of unchanged drug is a major route of elimination for 25%–30% of drugs administered to humans
 - During adulthood, there is a slow decline in renal function, about 1% per year
 - 2) Biliary and fecal excretion
 - 3) Other minor routes: Lung, breast milk, sweat, saliva, tear

Renal Excretion



Three distinct processes:

1) Glomerular filtration

- Glomerular filtration rate
- Extent of protein binding

2) Active tubular secretion

3) Passive tubular reabsorption

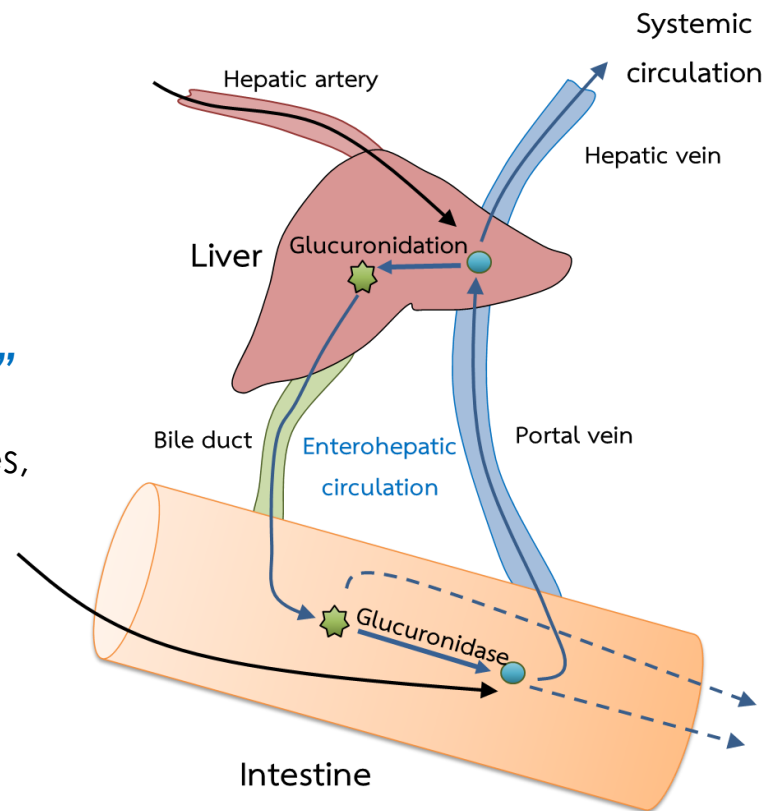
- Nonionized forms of weak acids and bases
- When urine is made more alkaline, weak acids are largely ionized and are excreted more rapidly and to a greater extent;
- Effects of changing urine pH are opposite for weak bases

• Renal impairment may lead to a reduction in drug excretion

- ↑ Serum creatinine (Scr)
- ↓ Creatinine clearance (CrCl) or eGFR

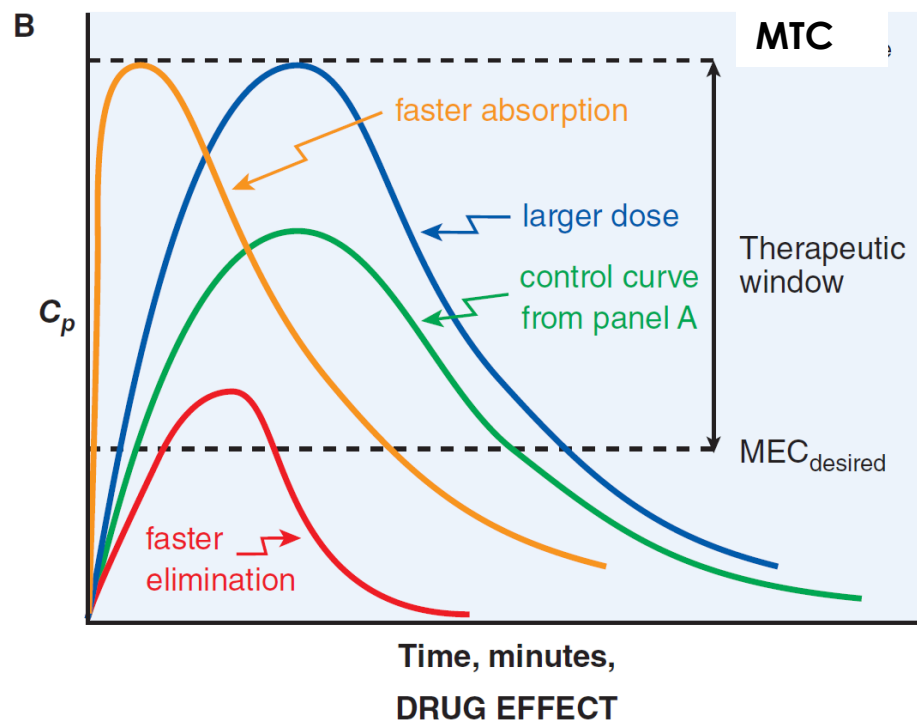
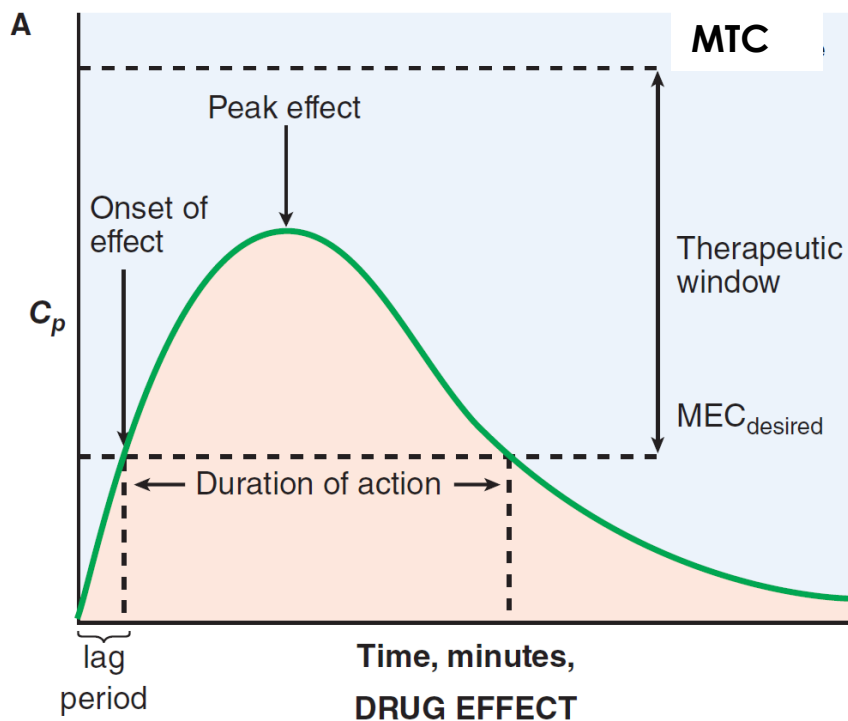
Biliary and Fecal Excretion

- Drugs and metabolites present in bile are released into the GI tract
- The drugs and metabolites can be reabsorbed into the body from the intestine called “**enterohepatic recycling**”
 - Conjugated metabolites such as glucuronides, may require enzymatic hydrolysis by the intestinal microflora
 - May prolong significantly the presence of a drug (or toxin) and its effects within the body
- Biliary excretions and unabsorbed drug are excreted in the feces



Adapted from: AANA J. 2020;88(1):71-6.

Understanding pharmacokinetic parameters



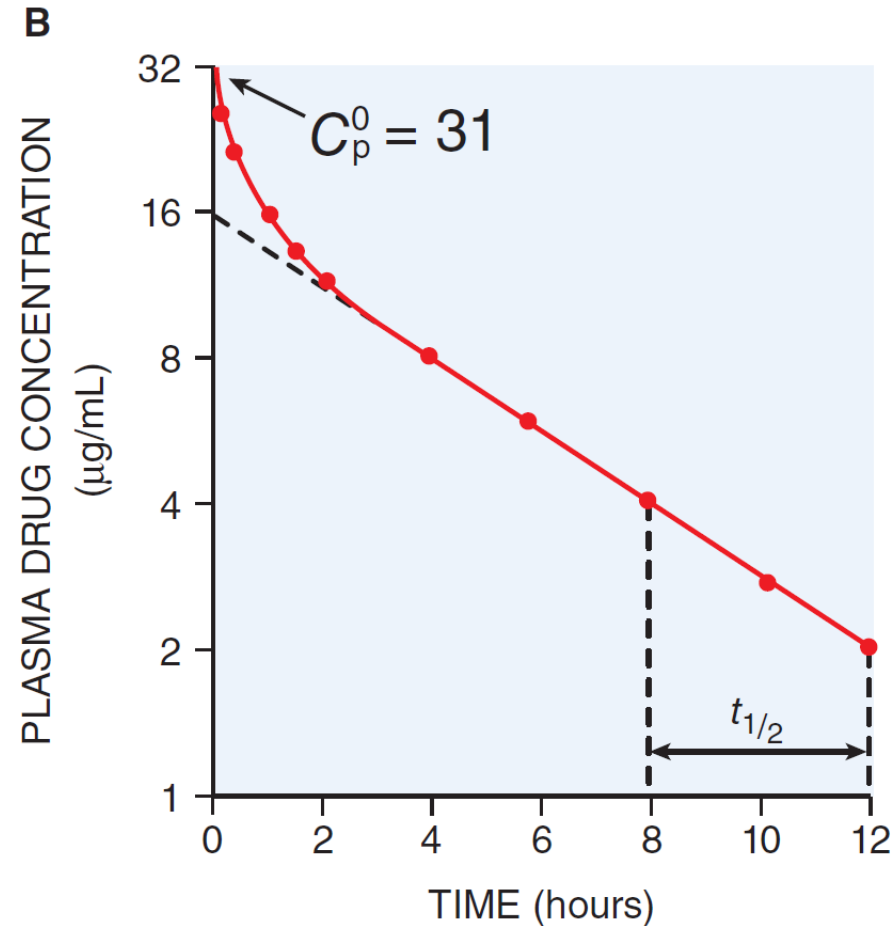
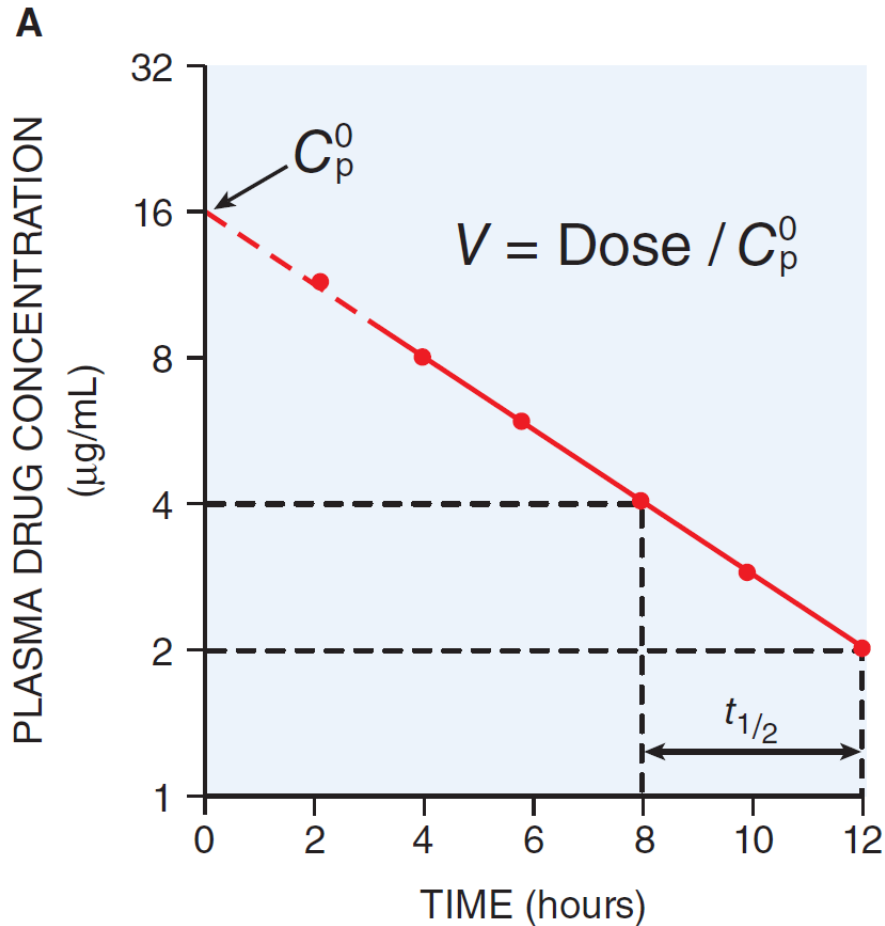
- C_p = Plasma concentration
- MEC = Minimum effective concentration
- MTC = Minimum toxic concentration



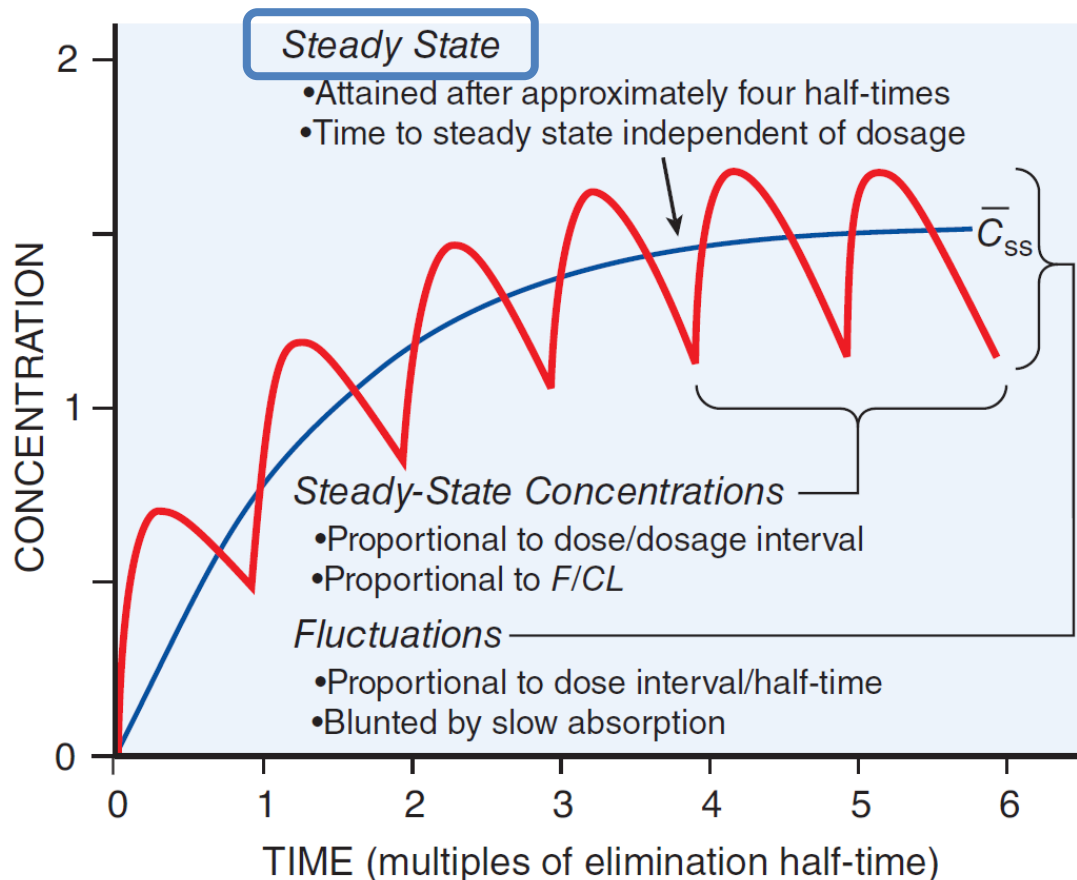
Understanding pharmacokinetic parameters (Cont.1)

Parameter	Name	Significance	Key features
C _{max}	Maximum plasma concentration	The highest drug concentration observed in plasma following administration	C _{max} and T _{max} are correlated and both depend on how quickly the drug enters into and is eliminated from the body
T _{max}	Time until C _{max} is reached	The time at which the highest drug concentration occurs	
AUC	Area under the concentration - time curve	The measure of the total systemic exposure to the drug	It represents the amount of unchanged drug that has reached the general circulation and it is useful to define the bioavailability of a drug
V _d	Volume of distribution	The apparent volume into which the drug is dissolved	It depends on binding to plasma proteins and tissues and it is useful to correlate the drug concentration in plasma with its amount in the body
t _{1/2}	Half-life in the terminal phase	The time taken for the plasma concentration to fall by one half once distribution equilibrium has been achieved	It is independent of the amount of drug in the body and it is useful for the determination of the frequency of drug administration
Cl	Clearance	The rate of drug elimination by all routes normalized to the concentration of the drug	It is the sum of all organs clearance, especially hepatic and renal clearance

Understanding pharmacokinetic parameters (Cont.2)



Understanding pharmacokinetic parameters (Cont.3)



Steady-state concentration (C_{ss}) is defined as the time during which the concentration remains stable or consistent when the drug is given repeatedly or continuously (IV infusion)

Other terms in pharmacokinetics

- **First-order kinetics**

- For most drugs in their therapeutic concentration ranges
- “The fraction of drug removed by metabolism is constant”
- The amount of drug metabolized per unit time is proportional to the plasma concentration of the drug (C_p)

- **Zero-order kinetics**

- For some drugs, such as ethanol, phenytoin and voriconazole
- “A constant amount of drug is metabolized per unit time”
- Metabolic capacity is saturated at the concentrations employed
- Zero-order kinetics can also occur at high (toxic) concentrations as drug-metabolizing capacity becomes saturated



Other terms in pharmacokinetics (Cont.)

- **Bioequivalence**

- Generic (local made) versus brand name (original)
- Drug products are considered to be **pharmaceutical equivalent** if they contain the same active ingredients and are identical in strength or concentration, dosage form, and route of administration
- Two pharmaceutically equivalent drug products are considered to be **bioequivalent** when the rates and extents of bioavailability of the active ingredient in the two products are not significantly different under suitable and identical test conditions
- Courts have not always found generic and brand name drugs to be legally equivalent

Thank you
Any Questions?



Post-test